Negative numbers and distance from zero

Explore and build the idea

20 daily questions include C1 and C2. Use separate paper when helpful.

Lesson focus: Order positive and negative numbers and describe their distances from zero.

R1. Find 3.45 + 5.078.

R2. Solve x + 11 = 18.

R3. A parallelogram has base 6, height 8, and slanted side 10 cm. Find its area.

R4. Three notebooks cost $2.45 each. What is the total cost?

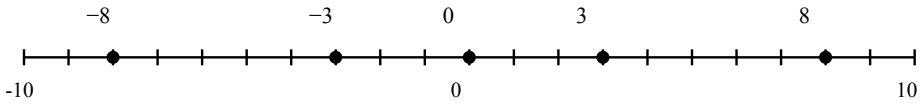
P1. Write a signed number for 4 degrees below zero, 7 m above sea level, and owing $9. State what zero means in each setting.

P2. Place −6, 1, and −2 on a number line. Which is least?

P3. Place −2.5, $-\frac{3}{4}$, and 0.5 on a number line. Order them from least to greatest.

P4. Find the opposite of −9. Then find the opposite of that result. Write −(−9).

Lesson model



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Practice and compare methods

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P5. Find $|-8|$, $|8|$, and $|0|$. Describe each as a distance.

P6. Compare -9 and -4 using $<$ or $>$. Compare their absolute values. Which temperature is warmer, $-9°C$ or $-4°C$?

F. Complete: -7 is __ units from zero, so $|-7|$ = __. Its opposite is __.

D. Ben says $-10 > -3$ because $10 > 3$. Use a number line to explain and repair his statement.

X1. Write signed numbers for 3 meters below sea level and 2 meters above it. State what zero represents.

X2. Order these values from least to greatest: $\frac{3}{2}$, -3, 0, -1.

X3. Find the opposite of -3, then its opposite. Explain why you return to the start.

X4. Find $|-3|$, $|3|$, and $|0|$. Compare the first two.

Worked example - compare with the slide model

Compare -1.5 and -0.5.

Locate zero, then mark both numbers to its left.

-1.5 lies one unit left of -0.5.

Therefore $-1.5 < -0.5$, but $|-1.5| > |-0.5|$.

Result: $-1.5 < -0.5$

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Challenge, explain and check

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C1. Three storage rooms are at −6°C, −1°C, and 4°C. Order them from coldest to warmest. Which is closest to 0°C? Find the temperature gap between the coldest and warmest rooms, and explain why the largest absolute value does not identify the warmest room.

C2. A diver is 12 m below sea level; a platform is 5 m above sea level. Write both signed locations. Find the distance from the diver to sea level and to the platform. Is the platform at the opposite of the diver's location? Explain.

Finish the lesson

S. The greater number lies __. The greater absolute value means __.

E. Order −6, 1.5, $-\frac{1}{4}$, and −2.5. Which number has the greatest absolute value? Give that absolute value.